Project: Unitree G1 Soccer Skills

CS2810 Teaching Team

1 Overview

This is the project component of Embodied AI CS2810 (Spring 2026). The task is to develop reinforcement learning skills for humanoid robot soccer using the Unitree G1 (29 dof) platform in `mjlab` physics simulation. The project centers on two tasks:

- **Shooting**: The G1 robot must approach and kick a stationary ball toward a goal. This is a perception-guided motor skill requiring motion tracking priors and ball trajectory prediction.
- **Goalkeeping**: The G1 robot must intercept an incoming ball launched along a parabolic trajectory. This is a reactive interception task requiring rapid end-effector positioning and ball trajectory estimation.

2 Project Phases

2.1 Phase 1: Basic Shooting and Goalkeeping (60% of project score)

Implement single-agent training for the shooter and goalkeeper tasks:

- Design observation spaces, reward functions, and neural network architectures following the designs in [1] and [2].
- Implement PPO training loops using the provided simulation environment and training entrypoint (`scripts/train.py`).
- Evaluate trained policies using the provided eval scripts. Target metrics (for full credit):
  - Shooter: 100% success rate over 50 trials (ball crosses goal plane inside the frame).
  - Goalkeeper: 100% block rate over 50 trials (ball does not enter the goal).

2.2 Phase 2: Adversarial Tournament (40% of project score)

Extend to adversarial play where a shooter and goalkeeper compete:

- Load two G1 robots simultaneously in the same MuJoCo scene.
- Implement self-play or league-based training to improve both shooting and goalkeeping against adversarial opponents.
- Compete in a class-wide tournament: each teams's shooter is evaluated against other teams' goalkeepers, and vice versa.

Important: Loading two G1 robots simultaneously for adversarial play is **not yet supported** in this template. It is required to extend the environment to support multi-robot scenes for Phase 2. This feature will be added in a future update.

3 Using the Template

The template is given at https://github.com/xiaojxkevin/G1-mjlab-soccer.

3.1 Repository Structure

The template provides:

- **Playable environments**: `Unitree-G1-Shooter` and `Unitree-G1-Goalkeeper` registered as mjlab tasks with configurable scene layout, ball physics, and MuJoCo visualization.
- **Evaluation scripts**: `eval_naive_shooter.py` and `eval_naive_goalkeeper.py` that load trained checkpoints, run headless batch trials, collect paper-matching metrics, and record videos. And you are allowed to modify these scripts if you want to use different observation spaces, reward functions etc.
- **Training configs**: Training environment designs under `config/training/` that specify observation spaces, reward functions, termination conditions, and domain randomization from the two papers. And it should be noticed that these are generated with coding agents and may contain errors. So you should carefully review and debug them before use.

3.2 Play and Visualization

Use the `scripts/play.py` script to visualize the environment:

```
# Shooter
python scripts/play.py Unitree-G1-Shooter --agent zero --viewer native

# Goalkeeper
python scripts/play.py Unitree-G1-Goalkeeper --agent zero --viewer native
```

3.3 Evaluation

Use the eval scripts to benchmark trained policies:

```
# Shooter
python scripts/eval_naive_shooter.py --headless --num-trials 50 \
    --checkpoint <path>

# Goalkeeper
python scripts/eval_naive_goalkeeper.py --headless --num-trials 50 \
    --checkpoint <path>
```

3.4 Implementing Training

The `train.py` script in `scripts/` serves as the training entrypoint. It is expected to register their own training tasks, implement the model classes, and run PPO training.

References

[1] J. Kong, X. Liu, Y. Lin, J. Han, S. Schwertfeger, C. Bai, and X. Li, "Learning soccer skills for humanoid robots: A progressive perception-action framework," 2026. [Online]. Available: https://arxiv.org/abs/2602.05310

[2] J. Ren, J. Long, T. Huang, H. Wang, Z. Wang, F. Jia, W. Zhang, J. Wang, P. Luo, and J. Pang, "Humanoid goalkeeper: Learning from position conditioned task-motion constraints," 2025.